RAJALAKSHMI ENGINEERING COLLEGE

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S2

Started on	Friday, 11 September 2026, 8:46 AM
State	Finished
Completed on	Friday, 11 September 2026, 9:19 AM
Time taken	32 mins 45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2		**g**	x	**t**	**x**	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int main() {
    char s1[1000], s2[1000];
    int dp[1000][1000];
    int i, j;

    scanf("%s",s1);
    scanf("%s",s2);

    int n = strlen(s1);
    int m = strlen(s2);

    for (i=0;i<=n;i++)
        dp[i][0] = 0;

    for (j=0;j<=m;j++)
        dp[0][j] = 0;

    for (i=1;i<=n;i++) {
        for (j=1;j<=m;j++) {
            if (s1[i-1]==s2[j-1])
                dp[i][j] = dp[i-1][j-1] + 1;
            else
                dp[i][j] = dp[i-1][j] > dp[i][j-1] ? dp[i-1][j] : dp[i][j-1];
        }
    }

    printf("%d\n", dp[n][m]);
    return 0;
}

```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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